

17.5

11)

$$\Delta x_1 = \frac{D\lambda}{d}$$

$$= \frac{1.50 \times 6.328 \times 10^{-7}}{3 \times 10^{-4}}$$

$$= 3.164 \text{ mm}$$

(12)

$$\Delta x_2 = \frac{D\lambda'}{d}$$

$$= \frac{D}{d} \cdot \frac{\lambda}{n}$$

$$= \frac{\Delta x_1}{n}$$

$$= 2.38 \text{ mm}$$

17.9.

$$n_1 \sin i = n_2 \sin r$$

$$\sin r = \frac{1 \times \sin 52^\circ}{1.4} \approx 0.5629$$

$$\cos r = \sqrt{1 - \sin^2 r} \approx 0.8205$$

$$2nd \cos r = k\lambda$$

$$2 \times 1.4 \times d \times 0.8205 = 1 \times 500 \text{ nm}$$

$$d \approx 216.1 \text{ nm}$$

17.10.

$$2nd = (k_1 + \frac{1}{2}) \lambda_1$$

$$\lambda_1 = 680 \text{ nm}$$

$$2nd = k_2 \lambda_2$$

$$\lambda_2 = 510 \text{ nm}$$

$$\therefore (k_1 + \frac{1}{2}) \lambda_1 = k_2 \lambda_2$$

$$(k_1 + 0.5) \times 680 = k_2 \times 510$$

$$k_1 = 1, k_2 = 2$$

~~$$2nd = k_2 \lambda_2$$~~

$$2 \times 1.33 \times d = 2 \times 510 \text{ nm}$$

$$d \approx 383.5 \text{ nm}$$

